

Multi-Agent Scientific Paper Review

qwen3:8b · May 20, 2026 12:12

File	REVISTA_Project.pdf
Words	6,252
Sections Detected	preamble, results, methodology, keywords, conclusions, experiments, references, title
Review Mode	DEEP
Model	qwen3:8b

■ **AI-ASSISTED REVIEW NOTICE:** This report was generated by CHVN Paper Reviewer, an AI-powered multi-agent system running locally. It is intended as a supplementary tool to assist human peer reviewers. All findings must be validated by a qualified domain expert before editorial decisions are made. The AI may miss context-specific nuances and should not be used as the sole basis for acceptance or rejection.

Parser Warnings

- Abstract not detected as separate section — content included in full text.

Editorial Decision

MAJOR REVISION**Weighted Score: 2.8 / 5.0**
Confidence: 80%

Executive Summary

The manuscript presents a study on the integration of IoT-based monitoring systems with solar-powered refrigeration for rural communities. It outlines a technically viable and economically feasible solution for off-grid applications.

Decision Rationale

The manuscript has significant strengths in its technical approach and relevance to sustainable development, but it requires major revisions to address critical issues such as the lack of a structured abstract, missing statistical analysis, and an inadequate discussion section. These revisions are necessary to improve the clarity, rigor, and completeness of the study.

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	5.0	Excellent	■■■■■
Manuscript Structure	2.0	Weak	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	1.0	Very Weak	■■■■■

Results & Evidence	3.0	Acceptable	■■■■■
Discussion & Conclusions	1.5	Very Weak	■■■■■
Figures & Tables	3.5	Good	■■■■■
Scientific Writing	3.0	Acceptable	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	1.2	Very Weak	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 5.0 / 5.0 | Confidence: 100%

Strengths:

- [TITLE] The title is specific and informative, clearly stating the purpose of the research and the technology used.
- [ABSTRACT] The abstract follows a structured format with clear background, objective, methods, results, and conclusions.
- [KEYWORDS] The keywords are comprehensive and include both domain-specific terms and general terms for discoverability.

Minor Comments:

- [TITLE] The title is slightly redundant due to the repetition of the Spanish and English versions. Consider shortening to improve readability.

Recommendations:

- [TITLE] Consider removing the English translation of the title to avoid redundancy and improve clarity.
-

StructureReviewer

Manuscript structure and logical coherence

Score: 2.0 / 5.0 | Confidence: 95%

Strengths:

- Clear title and keywords that align with the content
- Relevant keywords such as 'IoT monitoring' and 'photovoltaic system'

Weaknesses:

- Missing abstract section which is essential for summarizing the study
- Research problem not clearly stated in the introduction
- Research question or hypothesis not explicitly declared
- Objectives not stated and not addressed in the conclusions
- Section balance is uneven with the methodology and results being incomplete
- Transitions between sections are missing or unclear
- Keywords section contains incomplete sentences and is not self-contained
- References are incomplete and lack proper formatting
- Title is duplicated and not fully descriptive
- Results and methodology sections are incomplete and lack sufficient detail
- Conclusions are not supported by the data presented
- No statistical validity or reproducibility information is provided
- No ethical compliance or conflict-of-interest declaration is mentioned

Major Comments:

- The manuscript is missing a critical abstract section, which is essential for summarizing the study and guiding readers.
- The research problem is not clearly stated in the introduction, which undermines the clarity of the study's purpose.
- There is no explicit declaration of a research question or hypothesis, which is a fundamental element of scientific writing.
- The objectives of the study are not stated, and the conclusions do not address these objectives, leading to a disconnect between the stated goals and the final outcomes.
- The methodology and results sections are incomplete and lack sufficient detail to support the conclusions.

- The keywords section contains incomplete sentences and is not self-contained, which affects the searchability and clarity of the manuscript.
- The references are incomplete and lack proper formatting, which is a significant issue for academic integrity.
- The title is duplicated and not fully descriptive, which may confuse readers about the study's focus.
- The conclusions are not supported by the data presented, which weakens the validity of the findings.
- There is no mention of ethical compliance or conflict-of-interest declarations, which is required for scientific integrity.

Minor Comments:

- The title is duplicated and not fully descriptive.
- The keywords section contains incomplete sentences.
- The references are incomplete and lack proper formatting.

Recommendations:

- Add an abstract section to summarize the study and guide readers.
- Clearly state the research problem in the introduction.
- Declare the research question or hypothesis explicitly.
- State the objectives of the study and ensure they are addressed in the conclusions.
- Complete the methodology and results sections with sufficient detail to support the conclusions.
- Revise the keywords section to ensure it is self-contained and complete.
- Complete the references section with proper formatting and ensure all citations are included.
- Revise the title to be more descriptive and avoid duplication.
- Ensure the conclusions are supported by the data presented.
- Include a section on ethical compliance and conflict-of-interest declarations.

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: **4.0 / 5.0** | Confidence: **95%**

Strengths:

- The paper provides a clear and detailed description of the prototype design and construction.
- The thermodynamic analysis and COP calculations are well-documented and supported by equations.
- The integration of IoT monitoring is clearly explained and justified.
- The paper addresses real-world constraints such as rural energy poverty and lack of grid access.

Weaknesses:

- The methodology section is not clearly separated from the introduction and results, which may confuse readers about the logical progression of the research.
- Equation (1) is not explicitly defined in the text, though it is referenced in the text.
- The paper lacks a detailed description of the experimental setup, including the exact conditions under which the COP was calculated.
- The paper does not provide a detailed description of the data collection and analysis methods for the IoT monitoring system.

Major Comments:

- The methodology section is not clearly separated from the introduction and results, which may confuse readers about the logical progression of the research.

Minor Comments:

- Equation (1) is not explicitly defined in the text, though it is referenced in the text.
- The paper lacks a detailed description of the experimental setup, including the exact conditions under which the COP was calculated.

- The paper does not provide a detailed description of the data collection and analysis methods for the IoT monitoring system.

Recommendations:

- Separate the methodology section from the introduction and results to improve clarity and logical flow.
- Define all equations explicitly, including Equation (1), to ensure reproducibility.
- Provide a detailed description of the experimental setup, including the exact conditions under which the COP was calculated.
- Include a detailed description of the data collection and analysis methods for the IoT monitoring system.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: **1.0 / 5.0** | Confidence: 80%

Strengths:

- The manuscript provides detailed technical descriptions of the system and its components, which is valuable for engineering replication.
- The use of IoT for real-time monitoring is well described and aligns with the stated objectives of the project.
- The comparison of refrigerants (R-134a vs. R-290) is well-structured and includes relevant properties such as GWP, COP, and environmental impact.
- The manuscript includes a comprehensive list of variables and their definitions, which is important for clarity and reproducibility.

Weaknesses:

- The manuscript lacks any statistical analysis to compare the performance of different refrigerants or system configurations.
- No inferential statistics are used to assess the significance of differences in COP, energy consumption, or other performance metrics.
- The absence of confidence intervals and effect sizes limits the ability to assess the reliability and magnitude of observed differences.
- The sample size justification is missing, which is critical for evaluating the validity of the results.
- No statistical software or code is referenced, which is essential for reproducibility in engineering research.

Major Comments:

- The manuscript fails to report any statistical tests, which is a major limitation for inferential analysis. The results are purely descriptive and do not support any statistical claims.
- The absence of confidence intervals and effect sizes prevents the reader from assessing the precision and magnitude of the observed differences.
- The lack of sample size justification and multiple comparison corrections undermines the validity of any comparative analysis.
- The manuscript does not provide any information on reproducibility, such as seeds, runs, or variance metrics, which is critical for scientific rigor.
- The absence of statistical software or code references makes it impossible to replicate the results, which is a major flaw in the methodology.

Minor Comments:

- The coefficient of variation (CV) is mentioned but not interpreted in the context of variability or performance.
- RMSE, MAE, and R^2 are mentioned but not compared across different refrigerants or system configurations.
- The use of IoT for real-time monitoring is well described, but no statistical analysis is provided to assess its effectiveness or reliability.

Recommendations:

- The manuscript should include statistical tests (e.g., Kruskal-Wallis with post-hoc Wilcoxon tests) to compare the performance of different refrigerants or system configurations.

- Confidence intervals and effect sizes should be reported for all key metrics (e.g., COP, energy consumption, RMSE).
- A justification for the sample size (e.g., 30 runs) should be provided, including a power analysis if applicable.
- Multiple comparison corrections (e.g., Holm) should be applied when comparing multiple refrigerants or system configurations.
- Assumptions for any statistical test (e.g., normality, independence, homoscedasticity) should be verified and reported.
- The manuscript should include information on reproducibility, such as seeds, runs, or variance metrics.
- Bootstrap confidence intervals should be reported for nonparametric analysis, especially when dealing with stochastic systems.
- The coefficient of variation (CV), RMSE, MAE, and R^2 should be interpreted in the context of performance and variability.
- Hypervolume and spread indicators should be clearly defined and interpreted if used in the analysis.

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: **3.5 / 5.0** | Confidence: 62%

Strengths:

- 8 figure references were detected in the manuscript.
- 8 sequentially numbered equations were detected, suggesting a substantial mathematical formulation.

Weaknesses:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Minor Comments:

- The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.

Recommendations:

- Manually inspect captions, axes, units, table headers, and variable definitions in the final PDF.
-

ResultsReviewer

Results validity and empirical support

Score: **3.0 / 5.0** | Confidence: 70%

Strengths:

- Quantitative results such as $COP_{real} = 4.72$, $COP_{Carnot} = 6.28$, and $\eta_{II} = 75\%$ are clearly presented with equations and units.
- Comparison with baseline or state-of-the-art is included, such as the COP of R-134a vs. R-290 in Table 2.
- Thermodynamic properties and calculations are detailed, including equations for QH, COP_{real} , and energy consumption.
- Figures and tables are referenced and described, with some captions indicating their content.

Weaknesses:

- The results section is incomplete and lacks a clear conclusion or summary of findings.
- The phrase 'The results confirm the technical and economic feasibility of' is cut off and does not specify what is being confirmed.
- Negative results are not reported, despite the presence of a 'Limitations of results acknowledged' criterion.
- The results section does not clearly state whether the objectives were met or not, which is critical for evaluation.
- The results are not fully aligned with the introduction, which sets out the objectives of the study.

Major Comments:

- The results section is incomplete and lacks a clear conclusion or summary of findings. The phrase 'The results confirm the technical and economic feasibility of' is cut off and does not specify what is being confirmed.
- Negative results are not reported, despite the presence of a 'Limitations of results acknowledged' criterion.
- The results section does not clearly state whether the objectives were met or not, which is critical for evaluation.
- The results are not fully aligned with the introduction, which sets out the objectives of the study.

Minor Comments:

- The phrase 'The results confirm the technical and economic feasibility of' is cut off and does not specify what is being confirmed.
- The results section lacks a clear summary of findings and does not tie back to the stated objectives.
- Some figures and tables are referenced but not fully described in the text, which could hinder reproducibility.

Recommendations:

- Complete the results section by explicitly stating whether the objectives were met and what the results confirm.
- Report any negative results or limitations in the performance of the system, such as deviations from expected COP or energy consumption.
- Ensure that all figures and tables are fully described in the text, with clear captions that are self-contained.
- Clarify the alignment between the stated objectives in the introduction and the results presented.

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: **1.5 / 5.0** | Confidence: 90%

Strengths:

- The manuscript provides a clear context for the study by referencing existing literature on solar-powered refrigeration systems.

Weaknesses:

- The Discussion section lacks a comprehensive comparison with prior work, merely citing references without analyzing how the current study differs or builds upon previous findings.
- The Discussion does not address unexpected findings or limitations of the study, which is critical for a rigorous scientific discussion.
- The Conclusions section does not directly derive from the results; instead, it reiterates the introduction and lacks specific, quantified outcomes.
- The manuscript does not answer the original research question or hypothesis, as the study's objectives are not clearly defined or aligned with the conclusions.
- The manuscript overgeneralizes by making broad claims about the viability of solar-powered refrigeration systems without sufficient data or context to support such assertions.
- Future work directions are not specified or justified, which limits the contribution of the study to the field.

Major Comments:

- The Discussion section fails to engage with the existing literature in a meaningful way. It merely cites references without comparing the current study's findings with prior work.
- The manuscript does not address the limitations of the study, which is essential for a balanced and critical discussion.
- The Conclusions section is not derived from the results but rather reiterates the introduction, which undermines the scientific rigor of the manuscript.
- The study's hypothesis or research question is not clearly stated or addressed in the conclusions, which is a significant gap in the manuscript.

Minor Comments:

- The manuscript lacks specific, quantified conclusions, which weakens the impact of the findings.
- The future work section is absent, which is a common expectation in scientific manuscripts.

Recommendations:

- The Discussion section should include a detailed comparison with prior studies, highlighting how the current work advances or differs from existing research.
- The manuscript should clearly state the limitations of the study, such as sample size, geographic constraints, or technical challenges.
- The Conclusions section should be directly derived from the results, with specific, quantified outcomes and a clear alignment with the research question.
- The manuscript should define the original hypothesis or research question and ensure that the conclusions explicitly answer it.
- Future work directions should be concrete, justified by the study's findings, and aligned with the limitations and gaps identified.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.0 / 5.0** | Confidence: 55%

Strengths:

- The manuscript contains enough extracted text for a representative writing review.

Weaknesses:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Minor Comments:

- The model returned malformed JSON for writing review, so this fallback cannot provide sentence-level editing comments.

Recommendations:

- Review representative paragraphs manually for transition quality, undefined acronyms, and overly long sentences.
- Ensure technical terms are defined on first use and used consistently across sections.

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 70%

Strengths:

- A mix of recent and older references, with some relevant technical sources.
- Some references are from reputable institutions like FAO, World Bank, and NIST.

Weaknesses:

- The references section is inconsistent in citation style, with a mix of numbered and unnumbered citations.
- Many references are not properly formatted and lack standard citation elements (e.g., authors, journal names, page numbers).
- The references section contains a significant number of non-technical or irrelevant sources, such as government reports on poverty and food loss in Colombia.
- There is a lack of seminal works or foundational literature on refrigeration systems, thermodynamics, and photovoltaic systems.
- The references section lacks a balance between recent and classic works, with many older references from the 1980s and 1990s.

Major Comments:

- The references section is not properly formatted and lacks consistency in citation style. This undermines the credibility of the manuscript.

- The references section contains a significant number of non-technical or irrelevant sources, which do not support the claims made in the manuscript.
- The references section lacks seminal works or foundational literature on refrigeration systems, thermodynamics, and photovoltaic systems.
- The references section contains many outdated references from the 1980s and 1990s, which are not sufficient for a Q1 journal.

Minor Comments:

- Some references are not properly formatted and lack standard citation elements (e.g., authors, journal names, page numbers).
- The references section contains a mix of recent and older references, but the balance is not ideal for a Q1 journal.

Recommendations:

- Revise the references section to ensure consistent citation style (e.g., IEEE or APA).
- Include seminal works on refrigeration systems, thermodynamics, and photovoltaic systems.
- Remove non-technical or irrelevant references, such as government reports on poverty and food loss in Colombia.
- Ensure that all references are properly formatted with authors, journal names, page numbers, and publication years.
- Include more recent references from the last 5 years to meet the standards of a Q1 journal.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **1.2 / 5.0** | Confidence: 80%

Strengths:

- The manuscript presents a technically viable and economically feasible solution for rural communities, which is an important contribution to sustainable development.
- The integration of IoT-based monitoring systems with solar-powered refrigeration is a novel and relevant approach for off-grid applications.

Weaknesses:

- The manuscript lacks any mention of ethical approval, informed consent, or data availability, which are fundamental to research ethics and transparency.
- The limitations section is entirely absent, which is critical for a comprehensive evaluation of the study's scope and validity.
- The manuscript does not address reproducibility, which is essential for scientific validation and peer review.
- The conclusions are not supported by detailed data or statistical analysis, which undermines the credibility of the claims made.
- The methodology section is incomplete and lacks specific details on the experimental design, data collection, and analysis methods.

Major Comments:

- The manuscript fails to meet basic ethical standards by not disclosing ethical approval, informed consent, or data availability. These are required for any research involving human subjects or data collection.
- The absence of a limitations section is a major weakness, as it prevents readers from understanding the scope, validity, and generalizability of the findings.
- The manuscript does not provide sufficient detail on the experimental design, data collection, or analysis methods, which makes it impossible to assess the validity of the claims or replicate the study.
- The conclusions are not supported by detailed statistical analysis or data, which undermines the scientific rigor of the study.

Minor Comments:

- The keywords section is incomplete and does not fully reflect the content of the manuscript.
- The methodology section is too brief and lacks specific details on the implementation of the IoT-based monitoring system.

Recommendations:

- The authors should provide detailed information on ethical approval, informed consent, and data availability to meet the standards of Q1 journals.
- A comprehensive limitations section should be included to address the scope, validity, and generalizability of the findings.
- The methodology section should be expanded to include detailed descriptions of the experimental design, data collection, and analysis methods to ensure reproducibility.
- The conclusions should be supported by detailed statistical analysis and data to strengthen the scientific validity of the claims.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] The manuscript is missing a critical abstract section, which is essential for summarizing the study and guiding readers.
- [2] The research problem is not clearly stated in the introduction, which undermines the clarity of the study's purpose.
- [3] There is no explicit declaration of a research question or hypothesis, which is a fundamental element of scientific writing.
- [4] The objectives of the study are not stated, and the conclusions do not address these objectives, leading to a disconnect between the stated goals and the final outcomes.
- [5] The methodology and results sections are incomplete and lack sufficient detail to support the conclusions.
- [6] The keywords section contains incomplete sentences and is not self-contained, which affects the searchability and clarity of the manuscript.
- [7] The references are incomplete and lack proper formatting, which is a significant issue for academic integrity.
- [8] The title is duplicated and not fully descriptive, which may confuse readers about the study's focus.
- [9] The conclusions are not supported by the data presented, which weakens the validity of the findings.
- [10] There is no mention of ethical compliance or conflict-of-interest declarations, which is required for scientific integrity.
- [11] The methodology section is not clearly separated from the introduction and results, which may confuse readers about the logical progression of the research.
- [12] The manuscript fails to report any statistical tests, which is a major limitation for inferential analysis. The results are purely descriptive and do not support any statistical claims.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] [TITLE] The title is slightly redundant due to the repetition of the Spanish and English versions. Consider shortening to improve readability.
- [2] The title is duplicated and not fully descriptive.
- [3] The keywords section contains incomplete sentences.
- [4] The references are incomplete and lack proper formatting.
- [5] Equation (1) is not explicitly defined in the text, though it is referenced in the text.
- [6] The paper lacks a detailed description of the experimental setup, including the exact conditions under which the COP was calculated.
- [7] The paper does not provide a detailed description of the data collection and analysis methods for the IoT monitoring system.
- [8] The coefficient of variation (CV) is mentioned but not interpreted in the context of variability or performance.
- [9] RMSE, MAE, and R^2 are mentioned but not compared across different refrigerants or system configurations.
- [10] The use of IoT for real-time monitoring is well described, but no statistical analysis is provided to assess its effectiveness or reliability.
- [11] The model response could not be used reliably, so this fallback verifies presence and numbering but cannot judge visual quality or table body completeness.
- [12] The phrase 'The results confirm the technical and economic feasibility of' is cut off and does not specify what is being confirmed.

Specific Improvement Recommendations

1. [TITLE] Consider removing the English translation of the title to avoid redundancy and improve clarity.
2. Add an abstract section to summarize the study and guide readers.
3. Clearly state the research problem in the introduction.
4. Declare the research question or hypothesis explicitly.
5. State the objectives of the study and ensure they are addressed in the conclusions.
6. Complete the methodology and results sections with sufficient detail to support the conclusions.
7. Revise the keywords section to ensure it is self-contained and complete.
8. Complete the references section with proper formatting and ensure all citations are included.
9. Revise the title to be more descriptive and avoid duplication.
10. Ensure the conclusions are supported by the data presented.
11. Include a section on ethical compliance and conflict-of-interest declarations.
12. Separate the methodology section from the introduction and results to improve clarity and logical flow.